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Advanced Quality Control Workflow for Full-Size Core Images: Enhancing Machine Learning Training for Geological Applications

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Abstract

Machine learning (ML) has shown substantial promise in automating geological core description, supporting tasks such as lithology classification, porosity estimation, and texture prediction. However, the performance and reliability of ML models are heavily dependent on the quality and consistency of the input data. Core photographs, often collected under varied field conditions, frequently suffer from distortions such as color tint shifts, crushed or missing intervals, and the presence of non-geological visual artifacts like plastic wraps or labels. These issues reduce model accuracy and generalizability, particularly when applied across different fields or data collection settings.

This paper presents an advanced quality control (QC) workflow specifically designed for preprocessing full-size geological core images before their use in ML-based interpretation workflows. The system comprises three modular components: a Random Forest classifier for detecting tinted images based on RGB histogram statistics; a U-Net-based convolutional neural network (CNN) for segmenting crushed core intervals; and a MobileNetV2 CNN for identifying non-core elements. The workflow was trained and validated on available industrial datasets and demonstrated improvements in lithology prediction performance.

The QC system is fully modular, customizable, and suitable for integration into broader digital subsurface platforms. It enables scalable, reproducible filtering and standardization of geological imagery at field-wide scale, reducing manual review and improving downstream ML performance. This work introduces the novel fully documented and operational end-to-end ML-based QC solution tailored for full-size core images. This approach supports robust AI deployment in digital reservoir characterization, particularly in multi-well, multi-field contexts where data variability is high.

Introduction

Geological core description remains a fundamental component of subsurface reservoir characterization, informing interpretations of stratigraphy, lithology, and petrophysical properties. However, the conventional workflow for core analysis is time-intensive, subjective, and dependent on continuous expert involvement. This often results in inconsistent lithological logs, delays in project decisions, and underutilization of

valuable imagery—particularly in operations with extensive coring but limited onsite expertise. Given the high cost of core acquisition, the need for scalable and objective alternatives is increasingly urgent.

Machine learning (ML), especially deep learning models such as convolutional neural networks (CNNs), has shown significant potential in automating lithological classification and core photo interpretation (Baraboshkin et al., 2018; Abashkin et al., 2020; Alzubaidi et al., 2021). Recent advances include lithology classifiers trained on full-core imagery (Geng et al., 2020; Boiger et al., 2024), Bayesian neural networks for lithofacies segmentation (Xie et al., 2021), and multi-property regression models integrating core and well log data (Lin et al., 2021). Emerging work on foundation models such as DINOv2 (Brondolo and Beaussant, 2024) further underscores the growing generalizability of AI approaches in geoscientific image analysis.

Despite these advances, a persistent challenge remains: the quality and reliability of training data used for ML model development. Full-size core photographs frequently suffer from field-induced artifacts such as lighting distortions, crushed or missing intervals, plastic wraps, and non-geological labels. These inconsistencies introduce visual noise that degrades model performance and generalizability.

Several studies have recognized these limitations. For instance, Abashkin et al. (2020), Baraboshkin et al. (2022), and Tims et al. (2024) have highlighted issues stemming from image quality degradation. Industry applications (Datarock, 2023; SRK Consulting, 2024) stress the need for standardized image acquisition protocols but often fall short of providing formal QC procedures. McDonald (2021) and Yang (2024) further emphasized the negative impact of distorted images on lithology inference, advocating for more systematic preprocessing strategies. Tolstaya et al. (2023) additionally proposed semi-automated workflows to support dataset curation, though these remain underdocumented and limited in scope. In parallel, augmentation methods and manual filtering continue to be employed to improve data diversity (Baraboshkin et al., 2022); however, many existing workflows still rely on implicitly curated datasets, with little documentation of preprocessing standards (e.g., Lin et al., 2021; Yamada et al., 2024).

To bridge this gap, this study presents a modular, ML-based QC pipeline for full-resolution core images. The system incorporates statistical classification, segmentation, and object detection models to identify and remove color distortions, crushed intervals, and non-core elements. This ensures that downstream ML models are trained only on geologically relevant and visually consistent imagery. By formalizing the QC process, the proposed solution enhances dataset reliability, supports cross-field reuse, and improves robustness in core-based AI interpretation workflows.

Materials and Methods

The full dataset used for developing and validating the QC workflow consists of approximately 2,000 full-size RGB core photographs acquired from multiple boreholes. The images are captured in daylight conditions using a standardized setup and stored at high resolution. They represent a wide range of lithological intervals, rock textures, and visual conditions, including intact cores, fractured zones, and mechanically crushed sections.

Each image spans a continuous vertical interval and includes depth markings, borehole identifiers, and, in many cases, extraneous elements such as plastic wrap, handwritten labels, and artifacts from packaging or scanning (example in Fig. 1).



Figure 1—Example of full-size core photo.

The images used in this study reflect a typical operational setting for industrial-scale coring projects and serve as a representative benchmark for developing automated preprocessing solutions in ML-driven reservoir characterization pipelines. The QC pipeline for geological core image analysis integrates three

ML models: Random Forests (RF), U-Net-based CNNs, and MobileNetV2-based CNNs. Each algorithm is selected based on its suitability for a specific QC task, leveraging its architectural strengths and learning capabilities.

Tint Shift Detection Using Random Forest

The first stage of the QC workflow targets core images affected by abnormal color tint, often caused by lighting imbalance or channel oversaturation. These distortions degrade the reliability of downstream ML models by introducing color bias and inconsistency in extracted visual features.

ML Algorithm Background: Random Forest for Tint Detection. Random Forests, introduced by Breiman (2001), are ensemble learning methods that construct multiple decision trees during training and output the mode of the classes (classification) or mean prediction (regression) of the individual trees. The algorithm introduces randomness in two ways: by bootstrapping the data (sampling with replacement) and by selecting a random subset of features for splitting at each node. Given a dataset $D = \{(x_i, y_i)\}_{i=1}^n$, the RF algorithm builds T decision trees $\{h_t(x)\}_{t=1}^T$. For classification, the final prediction y for an input x is:

$$\hat{y} = \text{mode}\{h_t(x)\}_{t=1}^T \quad (1)$$

Illustrative diagram of the Random Forest algorithm is presented in Fig. 2.

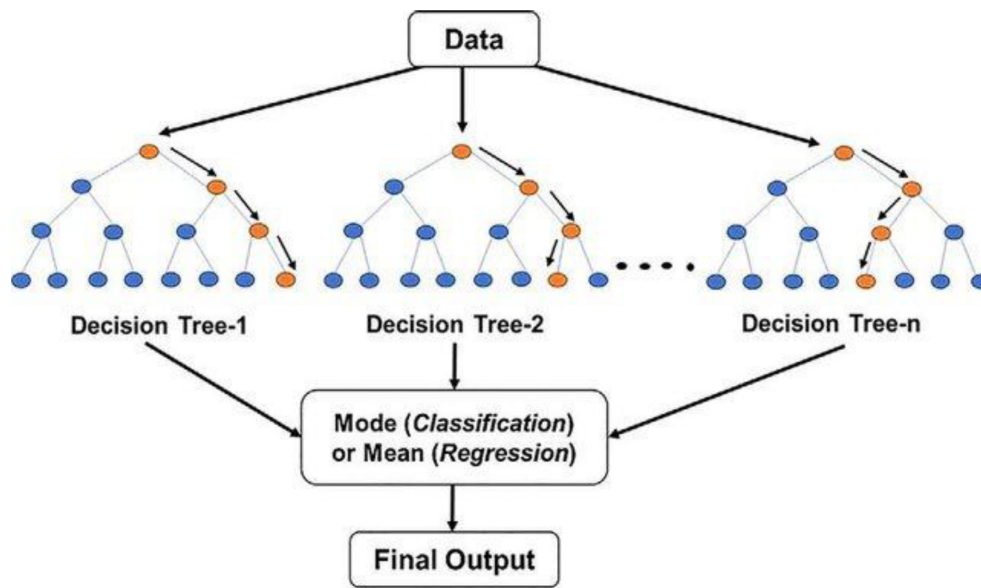


Figure 2—Schematic diagram of the Random Forest algorithm (Rani et al., 2022).

RF is well-suited for this task due to its ability to handle high-dimensional input and capture non-linear relationships between statistical image features and color distortions.

Application: Tint Shifting Detection Workflow. A binary RF classifier was trained to distinguish between valid and tinted core images. The dataset consisted of manually annotated examples labeled as either "good" or "tint-shifted".

Each image was converted into a 1,524-dimensional embedding using RGB channel-wise histograms and cumulative distribution functions. These features capture color balance, intensity distribution, and visual uniformity.

The RF classifier was trained using 40 estimators, a maximum tree depth of 6, and a minimum of 4 samples per leaf. Stratified 3-fold cross-validation was applied to ensure robustness to data imbalance. The

final trained model was then integrated into the QC pipeline for automatic detection and filtering of color-distorted images.

Crushed Core Segmentation Using U-Net

The second module in the QC pipeline addresses the problem of detecting crushed or fragmented intervals within full-size core images. Crushed sections lack coherent geological structure and may introduce misleading visual patterns during feature extraction for lithology classification. Therefore, their automatic segmentation and filtering is essential for ensuring the integrity of ML training data.

ML Algorithm Background: U-Net for Segmentation. U-Net, proposed by [Ronneberger et al. \(2015\)](#), is a convolutional neural network architecture designed for biomedical image segmentation. It consists of a contracting path (encoder) to capture context and a symmetric expanding path (decoder) that enables precise localization. Skip connections between corresponding layers in the encoder and decoder paths allow for the combination of low-level and high-level features.

The architecture applies a series of convolutional layers with ReLU activations, followed by max-pooling for downsampling in the encoder. The decoder uses transposed convolutions for upsampling, with concatenated skip connections that help recover spatial detail. A schematic illustration of the U-Net architecture is shown in [Fig. 3](#).

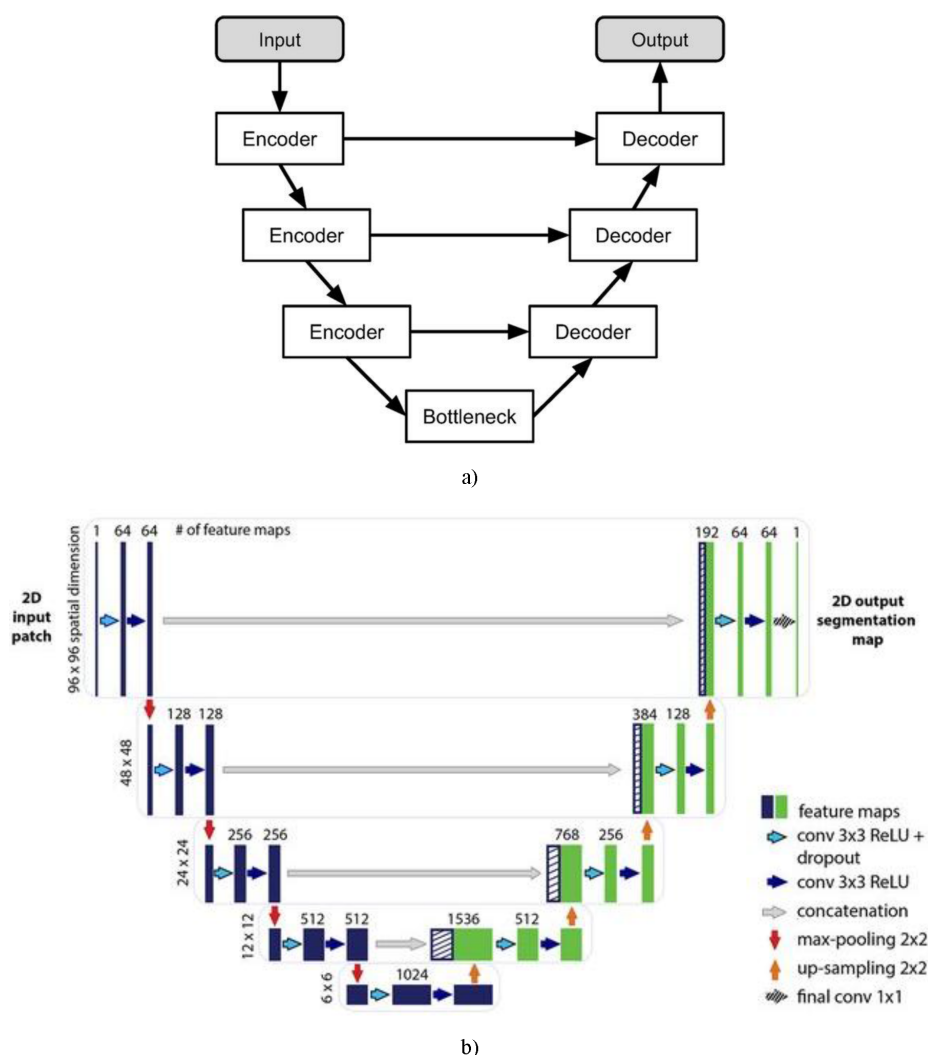


Figure 3—a) simplified representation ([Lee et al., 2022](#)); b) detailed illustration of the U-Net architecture ([Livne et al., 2019](#))

Crushed core regions in geological images present irregular textures and disrupted structures. U-Net's architecture is adept at capturing both the global context and fine-grained details necessary for segmenting such regions. By training on labeled datasets, the model learns to differentiate between intact and crushed core areas, enabling automated identification and exclusion of compromised sections.

Application: Crushed Core Segmentation Workflow. In the absence of manually labeled masks, a previously trained U-Net segmentation model is used to classify crushed and intact zones in the input images. The model generates binary masks and calculates the percentage of intact rock coverage for each image. To ensure data quality and model consistency, only images with an appropriate proportion of intact core are selected, excluding those dominated by crushed zones or noise. An example of a low-quality core photo is shown in Fig. 4. This thresholding step filters out irrelevant samples, reduces noise, and improves training stability.



Figure 4—Example of a low-quality core photo used for crushed core segmentation model training.

A refined U-Net model was then fine-tuned using a filtered dataset. The training process employed patch-based learning and dynamic cropping to improve generalization. Input patches were 256×256 pixels, with a magnification factor of 50 to increase variability by reusing full-size images with random crops. The model was trained using a batch size of 32 for 3 epochs.

The training used cross-entropy loss with label smoothing and class weights (foreground: 0.7, background: 0.3). The optimizer was AdamW with a weight decay of 1×10^{-4} . ReduceLROnPlateau scheduler was applied with a patience of 3, reduction factor of 0.5, and a minimum learning rate of 1×10^{-8} . Training was conducted in PyTorch Lightning with early stopping and checkpointing. Experiment tracking was handled via MLflow. To improve robustness, image augmentations such as horizontal and vertical flips, random 90° rotations, optional piecewise affine transformations, and random crop-resizing were applied during training. The final model was integrated into the QC workflow to automatically exclude images with excessive crushed rock content based on the calculated intact coverage threshold.

Non-Core Object Detection Using MobileNetV2

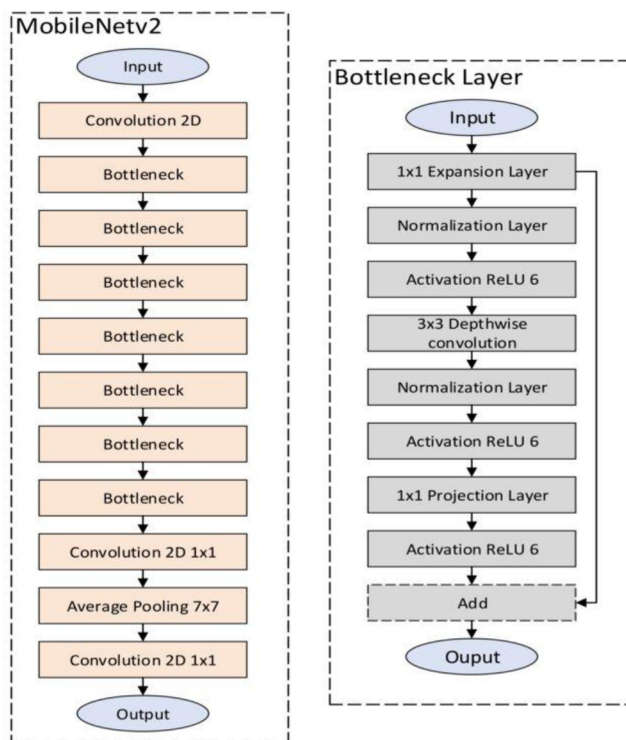
The third module in the QC pipeline is designed to detect and remove non-geological visual artifacts from core image patches. These may include plastic wraps, handwritten depth markers, printed labels, and measuring tape — all of which introduce visual noise that can degrade ML model accuracy and generalizability if left unfiltered.

ML Algorithm Background: MobileNetV2 for Lightweight Classification. MobileNetV2, developed by Sandler et al. (2018), is a lightweight CNN architecture optimized for mobile and embedded vision applications. It introduces two key innovations: inverted residuals and linear bottlenecks. The architecture uses depth-wise separable convolutions to reduce computational complexity while maintaining accuracy.

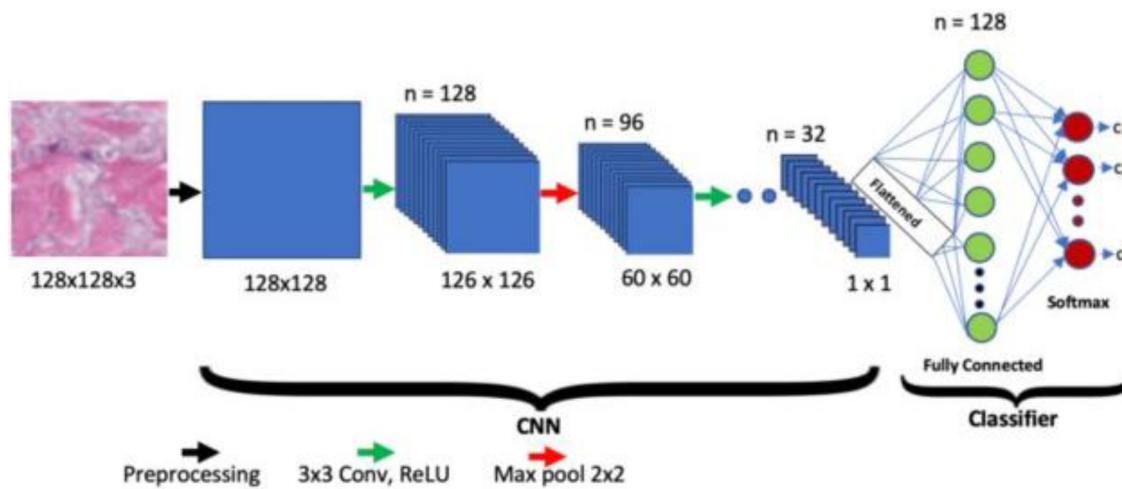
An inverted residual block in MobileNetV2 consists of:

1. A 1×1 convolution with ReLU6 activation (expansion layer).
2. A 3×3 depth-wise convolution with ReLU6 activation.
3. A 1×1 linear convolution (projection layer).

The residual connection is applied between the input and output of the block when the input and output dimensions match. Fig. 5 demonstrates the key architecture of the MobileNetV2 NN.



a)



b)

Figure 5—Architecture of the MobileNetV2 deep neural network: a) [Tragoudaras et al., 2022](#); b) by [Akay et al., 2021](#).

MobileNetV2's efficient architecture allows for real-time classification of image patches, distinguishing between core and non-core elements. Its balance between performance and computational efficiency makes it suitable for large-scale deployment in QC pipelines.

Application: Non-Core Patch Classification Workflow. To train the model, the QC-passed full-size core images were cropped into patches of 256×256 pixels using a depth-aligned sliding window approach. The

resulting dataset was semi-automatically labeled using a previously trained classifier, followed by expert review to verify and select representative samples of valid core and non-core content.

A MobileNetV2-based CNN was fine-tuned for binary classification using the following settings: Batch size: 64, Epochs: 10, Loss function: Cross-entropy. Optimizer: Adam, Learning rate scheduler: ReduceLROnPlateau, based on validation loss.

To improve generalization and reduce overfitting, standard image augmentations were applied during training, including horizontal and vertical flips, as well as 90° rotations. The model was implemented in PyTorch, and training was monitored using MLflow.

Integrated Pipeline Execution

After the independent training and validation of each module—tint detection, crushed core segmentation, and non-core patch classification—the models are sequentially integrated into a unified preprocessing pipeline. This pipeline operates on full-size core images and produces a cleaned, standardized dataset suitable for downstream ML applications, such as lithology classification or rock property prediction.

Workflow Overview. The pipeline consists of the following sequential steps:

1. **Input handling:** Full-size RGB core images are loaded from structured directories. Images are assumed to contain depth annotations and vary in quality and resolution.
2. **Tint filtering:** Each image is passed through the RF classifier trained on color distribution statistics. Images predicted as tint-shifted are flagged and excluded.
3. **Crushed zone filtering:** The remaining images are segmented using the fine-tuned U-Net model. A binary mask is generated per image, and intact rock coverage is calculated. Images below a defined threshold are removed from further processing.
4. **Patch generation and filtering:** Valid full-size images are cropped into 256×256 patches using a depth-aligned sliding window. These patches are classified using the MobileNetV2 model. Non-core patches—those containing plastic, labels, or measurement artifacts—are filtered out.
5. **Dataset output:** The final output includes cleaned full-resolution images, valid cropped image patches, binary masks (for reference or reuse), metadata logs documenting filtered images and classification decisions.

Implementation Details. The entire pipeline is implemented using Python with PyTorch, Scikit-learn, and PyTorch Lightning libraries. Execution is modular and notebook-based, allowing each model to be retrained or replaced independently. Configuration parameters (e.g., threshold values, batch sizes, model checkpoints) are customizable for different datasets or project conditions. The pipeline was validated on a real-world core image dataset, demonstrating robust scalability and compatibility with digital core analysis platforms. Its modular design supports integration into lithology prediction GUIs and other geoscientific ML frameworks.

Results and Discussions

Tint Shift Detection Results

The tint filtering model was evaluated using a stratified 3-fold cross-validation strategy on the labeled dataset. It achieved a mean accuracy of 0.99, with precision and recall ranging from 0.96 to 1.00, confirming high classification reliability.

When applied to the full dataset, the model identified and excluded 8.6% of images affected by color distortions, such as red/blue channel dominance and uneven lighting. Example of classified images are shown in Fig. 6. This filtering step improved visual uniformity across the dataset and ensured that downstream lithology classification was not biased by color artifacts.

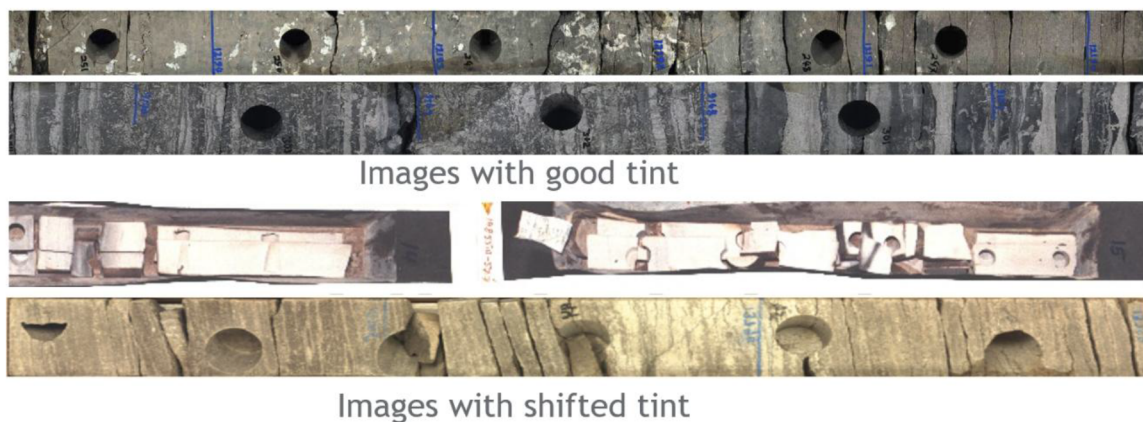


Figure 6—Examples of images with valid and shifted tint.

Crushed Core Segmentation Results

The U-Net-based crushed core segmentation model was evaluated on a held-out test set using standard semantic segmentation metrics. It achieved a Dice coefficient of 0.84, F1-score of 0.91, IoU of 0.84, precision of 0.84, and recall of 1.00, indicating strong and consistent performance across varying image conditions.

Using the model's binary masks, the intact core coverage was computed for each image. A filtering threshold of 70% was applied to exclude images with insufficient intact rock area. 9% were filtered due to high crushed content.

Visual examples of segmented images with different crushed core content are shown in Fig. 7. This segmentation-based filtering ensured that subsequent model training was focused only on images with sufficient geological information. Initial lithology classification experiments conducted with and without this filtering step suggest that excluding crushed intervals contributes to better texture feature stability and fewer classification artifacts.

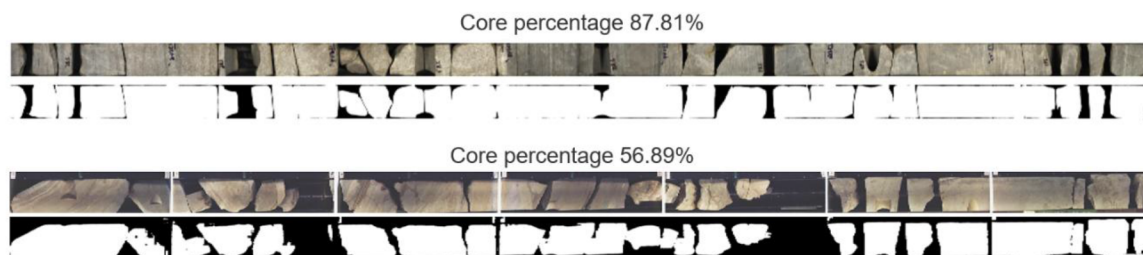


Figure 7—Examples of masks obtained from crushed core segmentation model.

Non-Core Object Detection Results

The MobileNetV2-based classifier was evaluated on a held-out test set to assess its ability to distinguish between valid core patches and those containing non-geological visual artifacts such as plastic wrap, printed labels, or depth markers. The model achieved a precision of 0.93, recall of 0.84, and F1-score of 0.88, demonstrating reliable and consistent performance in detecting irrelevant visual content.

Visual examples of classified rock and non-rock patches are shown in Fig. 8. The use of this filtering step significantly reduced noise in the training dataset, helping to ensure that downstream lithology and rock property models are trained on geologically relevant content.

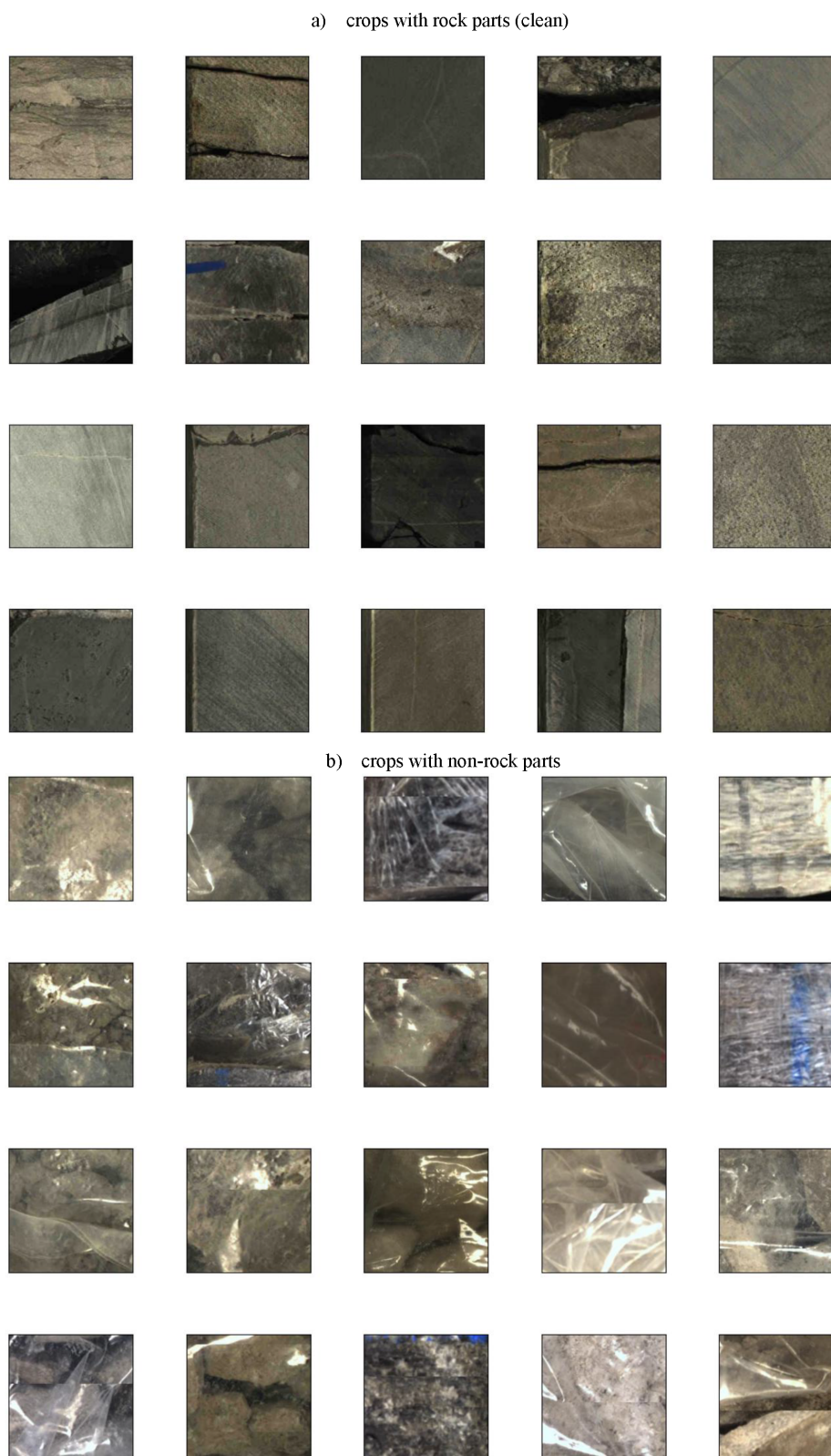


Figure 8—Prediction examples from the developed AI model for non-core parts detection.

Integrated QC Workflow Impact on Lithology Prediction

To evaluate the practical effect of the full QC workflow, a lithology classification model was trained on two datasets: one processed using the proposed QC pipeline and one based on raw, unfiltered core images. Both models used identical architectures, training configurations, and target labels to ensure comparability.

Preliminary experiments indicate that the model trained on the QC-filtered dataset achieved improved performance across key evaluation metrics. Validation accuracy, precision, and recall each increased by approximately 2–3%, suggesting enhanced generalization and more consistent predictions. Further testing on larger and more diverse datasets is expected to provide stronger evidence of this trend.

Conclusions

This study presents a comprehensive quality control (QC) workflow for filtering full-size geological core photographs prior to their use in ML-based geoscientific interpretation. The system integrates Random Forest, U-Net, and MobileNetV2 models to detect and filter common visual inconsistencies—such as tint shifts, crushed intervals, and non-core artifacts—that typically undermine ML model accuracy and generalizability. By addressing image quality at the source, the workflow improves dataset consistency, enhances training reliability, and reduces dependency on manual curation. While most prior efforts either overlook quality control entirely or rely on undocumented manual filtering and augmentation strategies, this study introduces the first fully documented, end-to-end QC pipeline specifically tailored to full-size geological core imagery.

Designed for configurability and field-scale deployment, the workflow:

- Streamlines image preprocessing with minimal expert intervention
- Improves the robustness of downstream ML models
- Enables wider reuse of historical datasets through standardized QC
- Facilitates integration into digital reservoir workflows at scale

Future work will further expand its application to broader and more diverse image archives, enhancing its adaptability and robustness for industrial-scale AI deployment in geoscience.

Nomenclature

CNN	= Convolutional Neural Network
D	= Dataset composed of image-feature and label pairs
$F1$	= F1-score, the harmonic mean of precision and recall
$h_t(x)$	= Prediction of the t -th decision tree in the Random Forest
IoU	= Intersection over Union, a metric for overlap between predicted and true segmentation masks
ML	= Machine Learning
$\text{mode}\{h_t(x)\}_{t=1}^T$	= MMajority voting rule for classification
n	= Total number of samples in the dataset
QC	= Quality Control
ReLU	= Rectified Linear Unit
RF	= Random Forest
RGB	= Red, Green, Blue (color channels)
T	= Total number of decision trees in the Random Forest
U-Net	= A specific CNN architecture for image segmentation
$\hat{y}$	= Predicted class label from the Random Forest model

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